



Modelling Financial Markets

Chapter 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ ETF: Exchange-Traded Fund
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-MIDAS: GARCH with Mixed Data Sampling (long-run component)
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH

Acronyms used in this material (2/2)

- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile–Quantile (plot)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

Outline

Motivation

From ARCH to GARCH

Estimation and Model Choice

Worked Example and Results by Market

Asymmetric Volatility

Innovations and Diagnostics

Forecasting and Forecast Evaluation

Extensions and Limits

AI for Scientific Discovery

Summary

References

Volatility Comes in Bursts



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Crashes are not confined to isolated days: large moves are followed by large moves
 - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- Question: can we turn the clustering into a **forecast** of tomorrow’s risk, and conduct valid inference on it?

Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4
BET	Bucharest Stock Exchange	2000	6 670	249.8
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3
EUR/RON reference rate	BNR	2005	5 352	252.4
Gold (XAU/USD)	spot market	2000	6 922	259.2

- Daily closes up to 18 September 2026; returns $r_t = 100 (\ln P_t - \ln P_{t-1})$, in %
- Equity indices and gold: weekdays only; gold volatility is annualised using its actual observation frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

Prerequisites Recap: Clustering and Heavy Tails (Chapter 1)

- Heavy tails: excess kurtosis S&P 500 10.7, BET 11.1, Bitcoin 12.0 (0 for the Normal distribution)
 - ▶ Annual volatility: Bitcoin 66.6%, S&P 500 19.2%, BET 22.1%
- S&P 500: lag-1 autocorrelation of r_t is -0.099, of r_t^2 is 0.313
 - ▶ All 50 autocorrelations of r_t^2 up to lag 50 lie outside ± 0.024 ; still 0.053 at lag 50
- The sign of tomorrow's return is almost unpredictable; its size is not: the object to model is $\text{Var}(r_t \mid \mathcal{F}_{t-1})$
- Charts of both facts: Quantlet below and Chapter 1

 MFM_ch5_arch_effects

Testing for ARCH Effects

- ▣ **McLeod–Li** [McLeod & Li \(1983\)](#): Ljung–Box $Q(10)$ on r_t^2 , χ_{10}^2 under no clustering
- ▣ **Engle's LM test** [Engle \(1982\)](#): nR^2 from regressing $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$, χ_5^2 ; $\hat{\varepsilon}_t = r_t - \bar{r}$
- ▣ The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
 - ▶ [Lee & King \(1993\)](#): one-sided score test built on the sum of the score components, $N(0, 1)$ under H_0 , rejects in the upper tail only
- ▣ Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
 - ▶ The ARCH(q) LM statistic is also the score test against GARCH(p, q) [Lee & King \(1993\)](#): the same statistic serves both alternatives

ARCH Effects in Six Markets

Series	Vol. (% p.a.)	Excess kurt.	$\hat{\rho}_1(r^2)$	$Q(10)$ on r^2	LM(5)
S&P 500	19.2	10.7	0.313	6133	1684
BET	22.1	11.1	0.330	2440	982
BET-TR	15.4	17.8	0.171	714	371
Bitcoin	66.6	12.0	0.137	213	115
EUR/RON	4.4	15.1	0.342	2420	871
Gold	17.6	7.7	0.110	744	329

- 5% critical values: χ_{10}^2 : 18.3; χ_5^2 : 11.1; every statistic is far above them (all p-values < 0.001)
- Clustering is universal: equities, crypto, a managed exchange rate and gold
- Weakest (but still decisive) for Bitcoin and gold, strongest for the S&P 500

The ARCH Model of Engle (1982)

- Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
 - ▶ Conditional variance $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$ ($\mathcal{F}_{t-1}$: information at $t-1$); unconditional $\sigma^2 = \mathbb{E}\sigma_t^2$
- **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2$ [Engle \(1982\)](#)
 - ▶ $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ARCH(1): unconditional variance $\sigma^2 = \omega/(1 - \alpha_1)$, finite if $\alpha_1 < 1$
 - ▶ Normal z_t : kurtosis $3(1 - \alpha_1^2)/(1 - 3\alpha_1^2)$, finite only if $3\alpha_1^2 < 1$
 - ▶ It equals 3 at $\alpha_1 = 0$, exceeds 3 for $0 < \alpha_1 < 1/\sqrt{3}$ and is infinite for $\alpha_1 \geq 1/\sqrt{3}$: heavy tails even with Normal innovations

Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

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Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC

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- ▣ Nobel Prize 2003: Engle “for methods of analyzing economic time series with time-varying volatility (ARCH)”
 - Shared with Clive Granger (cointegration)
- ▣ Nobel lecture: Engle (2004), on risk and volatility in financial practice
- ▣ Tim Bollerslev, Engle’s doctoral student, generalised ARCH to GARCH in 1986

From ARCH to GARCH (Bollerslev, 1986)

- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH(1,1)** [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0, \alpha \geq 0, \beta \geq 0$; α : reaction to news; β : memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
 - ▶ An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p, q) adds more lags; in practice GARCH(1,1) is the workhorse [Hansen & Lunde \(2005\)](#)

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$
 - ▶ An ARMA(1,1) in ε_t^2 with autoregressive coefficient $\alpha + \beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
 - ▶ $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
 - ▶ S&P 500, GARCH(1,1) with Normal innovations: $\rho_1 = 0.356$, $\rho_{10} = 0.298$; sample 0.313 and 0.248
- $\alpha + \beta$ is therefore called the **persistence** of volatility

Stationarity and the Unconditional Variance

- **Covariance stationarity:** $\alpha + \beta < 1$; then
 - ▶ $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$
- Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite
 - ▶ Yet the process is still **strictly** stationary: the condition is $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ [Nelson \(1990\)](#)
- Variance targeting: fix $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$ with the sample variance $\hat{\sigma}^2$

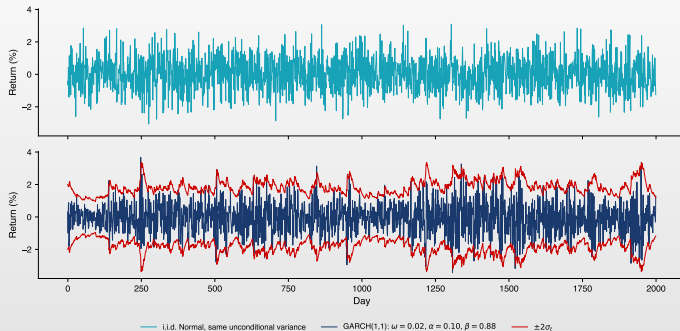
Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$
 - ▶ The gap to the long-run variance shrinks by the factor $\alpha + \beta$ each day
- **Half-life:** number of days after which half of a variance shock has gone
 - ▶ $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples: $\alpha + \beta = 0.95$: 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in $\alpha + \beta$ near 1 imply very different half-lives: estimation uncertainty matters (Seminar 5)

Fourth Moment and Kurtosis

- With Normal z_t , the kurtosis of ε_t is $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$
 - ▶ Finite only if $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (Seminar 1, Exercise A9)
- S&P 500, GARCH(1,1) with Normal innovations: $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$
 - ▶ $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.990435$: just below 1, so $K = 12.1$; sample kurtosis of the returns: 13.7
- K is ill-conditioned near the moment boundary: the denominator is $1 - 0.990435$
 - ▶ Rounding to $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$ gives 0.9916 and $K = 13.5$: report such functions from unrounded estimates
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t (Section 6)

Simulation: GARCH Produces Clustering



□ Top: i.i.d. Normal returns; bottom: GARCH(1,1) with the same unconditional variance (1) and $\pm 2\sigma_t$ bands

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Interpretation: Clustering from a Three-Parameter Rule

- Same unconditional variance, very different paths: quiet stretches and bursts appear only in the GARCH path
- On a long simulated path (one million days)
 - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2 : 0.26
 - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters
- Normal innovations, yet fat-tailed returns: the mixture of variances creates the tails

EWMA: a GARCH Special Case

- **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data and zero-mean returns
 - ▶ Weights $(1 - \lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
- EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: no parameter to estimate
- Price of simplicity: no mean reversion
 - ▶ At the close of day t , the forecast for any horizon equals the updated one-step variance σ_{t+1}^2 ; stationary GARCH pulls it towards σ^2
- We use EWMA as a benchmark in the forecast comparison of Section 7

Recap: ARCH and GARCH

Key points

- ▣ σ_t^2 is the variance given yesterday's information; it moves with the size of recent shocks
- ▣ $\text{GARCH}(1,1) = \text{ARCH}(\infty)$ with three parameters; $\alpha + \beta$ is the persistence
- ▣ $\alpha + \beta < 1$: finite long-run variance $\omega/(1 - \alpha - \beta)$ and half-life $\ln 0.5 / \ln(\alpha + \beta)$
- ▣ Clustering creates fat tails, even with Normal innovations
- ▣ EWMA is the IGARCH special case with $\omega = 0$

Maximum Likelihood

- ▣ Parameters $\theta = (\mu, \omega, \alpha, \beta)$; given θ , the recursion produces $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- ▣ Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$
 - ▶ Each day contributes the density of ε_t given the past: the likelihood factorises in time
- ▣ No closed form: numerical optimisation under $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- ▣ Start-up: σ_1^2 is set to the sample variance (its effect fades at rate β)
- ▣ Scale: returns in %, so that the variance is of order 1
 - ▶ EUR/RON (daily s.d. about 0.28%) is fitted on 10 r_t : unscaled, the optimiser stopped at a local optimum

QMLE: Consistency and Asymptotic Normality

- **Gaussian QMLE:** maximise the Normal likelihood even if z_t is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- **Consistency** (GARCH(1,1)) [Berkas, Horváth & Kokoszka \(2003\)](#), [Francq & Zakoïan \(2004\)](#)
 - ▶ Strict stationarity $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$; θ_0 in a compact set; z_t^2 non-degenerate
 - ▶ No moment condition on ε_t : IGARCH and infinite-variance cases are covered
- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E} z_t^4 < \infty$
 - ▶ $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \rightarrow N(0, (\kappa_z - 1)J^{-1})$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$; an estimated mean adds the score term $\varepsilon_t / \sigma_t^2$ and changes this form
 - ▶ Strict stationarity allows $\alpha + \beta \geq 1$ (IGARCH: [Lumsdaine \(1996\)](#)); explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) \geq 0$: [Jensen & Rahbek \(2004\)](#) cover (α, β) with ω fixed, as ω is not identified
- Sandwich $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$; the inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
 - ▶ A: minus the expected Hessian; B: variance of the score; robust / classical s.e. = $\sqrt{(\kappa_z - 1)/2}$, 1 under Normality

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
 - ▶ Theory: robust / classical s.e. = $\sqrt{(4.73 - 1)/2} = 1.37$; observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t gives $\hat{\nu} = 3.18 < 4$: under that law $\mathbb{E}z_t^4 = \infty$, and the condition for Gaussian-QMLE normality fails
 - ▶ Residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample; s.e. ratios 2.51 and 2.01
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$, to a non-Gaussian limit: its robust t -statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014), or the Student-t MLE (bounded score: no fourth moment needed if the law is right), with bootstrap inference near bounds

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Heavy-Tailed Innovations

- **Student-t** [Bollerslev \(1987\)](#): $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$, unit variance for $\nu > 2$
 - ▶ ν (degrees of freedom): the smaller ν , the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter $\lambda \in (-1, 1)$
 - ▶ $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
 - ▶ $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

Testing on the Boundary of the Parameter Space

- $LR = 2(\ell_1 - \ell_0) \rightarrow \chi_m^2$ only for interior restrictions (e.g. GJR vs GARCH: $\gamma = 0$, $m = 1$)
 - ▶ Non-nested models: **AIC** = $-2\ell + 2k$, **BIC** = $-2\ell + k \ln n$ (k parameters); lower is better
- GARCH restrictions $\alpha, \beta \geq 0$, $1/\nu \geq 0$, $\alpha + \beta \leq 1$ put many nulls **on the boundary**
 - ▶ One parameter on the boundary, regular likelihood, other parameters identified and interior: $LR \rightarrow \frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [Self & Liang \(1987\)](#), [Andrews \(2001\)](#); 5% critical value 2.71, not 3.84
 - ▶ Cases: the Normal distribution vs the Student-t distribution ($1/\nu = 0$); GJR with $\alpha = 0$ ($\gamma > 0$); IGARCH
 - ▶ Not covered: no ARCH ($\alpha = 0$ leaves β unidentified: a non-standard limit [Andrews \(2001\)](#)) and several active bounds at once
- Wald t -tests of a parameter estimated at the bound (e.g. $\hat{\alpha} = 0$) are not $N(0, 1)$
 - ▶ The parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary [Andrews \(2000\)](#): simulate under H_0 instead

Boundary Tests on Our Data

- Normal vs Student-t innovations, GARCH(1,1): $LR = 311.0$ (S&P 500), 1082.0 (Bitcoin)
 - ▶ Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- GJR-t, S&P 500: $\hat{\alpha} = 0.000$; its t -test is not standard, and the χ_1^2 law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$
- IGARCH, $H_0: \alpha + \beta = 1$ (GARCH-t): $LR = 1.11$ (S&P 500), 1.14 (BET)
 - ▶ S&P 500: $p = 0.29$ (χ_1^2), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 , $B = 999$)
 - ▶ BET: $p = 0.28$, 0.14 and 0.17
- Under H_0 , 52% of the re-estimates sit at $\alpha + \beta = 1$, as the mixture predicts: a unit root in variance cannot be rejected for either index

S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust s.e.	Classical s.e.	Ratio
ω	0.0157			
α	0.123	0.011	0.010	1.06
β	0.872	0.010	0.010	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Daily returns in %, 2000–2026; mean $\hat{\mu} = 0.079\%$ per day
- Half-life 131 days; long-run volatility $\sqrt{251.4 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$ p.a.

 MFM_ch5_garch_estimation

Interpretation: S&P 500 Estimates

- ▣ $\hat{\beta} = 0.872$: most of today's variance is yesterday's variance; $\hat{\alpha} = 0.123$: news adds a moderate update
- ▣ Persistence 0.9947: shocks to volatility last months, not days
- ▣ The long-run volatility (27.3%) exceeds the sample volatility (19.2%)
 - ▶ Near $\alpha + \beta = 1$, $\omega/(1 - \alpha - \beta)$ divides by a tiny number and is imprecise
- ▣ $\hat{\nu} = 6.26$: innovations clearly heavier-tailed than the Normal distribution
- ▣ Robust standard errors are about 5–7% larger than the classical ones for α and β

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0, 1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

- ΔBIC : BIC minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: an asymmetric variance equation and a heavy-tailed, skewed innovation
- Moving from Normal to Student-t innovations lowers BIC by about 300; adding asymmetry lowers it by a further 230–270
- Winner: EGARCH with skewed-t innovations; Sections 5–6 explain why

 MFM_ch5_garch_estimation

Recap: Estimation

Key points

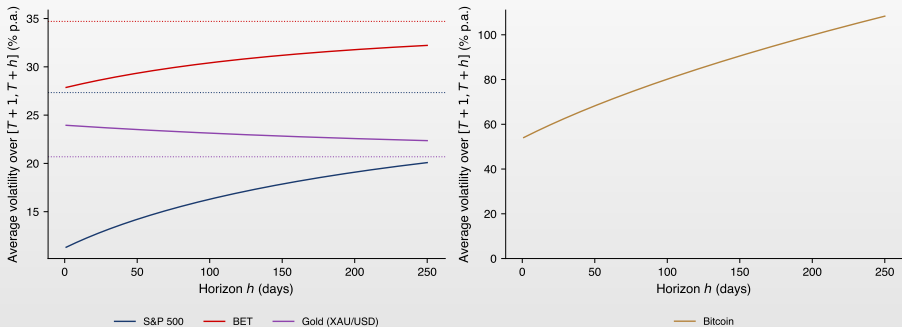
- ▣ GARCH is estimated by maximum likelihood; the likelihood is built day by day from the recursion
- ▣ Gaussian QMLE is consistent without moment conditions; its $\sqrt{n}$ normality needs $\mathbb{E}z_t^4 < \infty$, which fails for Bitcoin under the fitted Student-t law
- ▣ Student-t, skewed-t and GED capture the tails that GARCH alone leaves
- ▣ Nulls on the boundary ($1/\nu = 0$; GJR $\alpha = 0$): LR against $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$; IGARCH: simulate under H_0 ; no ARCH (β unidentified) is non-standard
- ▣ Rescale low-volatility series; compare non-nested models with AIC/BIC

Worked Example: One-Day and Ten-Day Forecasts

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}_T^2 = 0.565$
 - ▶ $\hat{\sigma}_{T+1}^2 = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_T^2 + \hat{\beta}\hat{\sigma}_T^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 2.97$; day 10: $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$
 - ▶ $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$, volatility 2.38%
 - ▶ The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}_{T+1}^2 < \hat{\sigma}^2$ and volatility is forecast to rise

 MFM_ch5_garch_estimation

Volatility Term Structure



Forecast from 18.09.2026 of the average volatility over the next h days (GARCH(1,1)-t); dotted: long-run level

Interpretation: Mean Reversion or Not

- ▣ S&P 500: from 11.3% (next day) to 20.1% (one year), slowly rising towards the long-run level
- ▣ Gold: today above its long run, so the curve declines, from 23.9% to 22.4%
- ▣ Bitcoin and EUR/RON: $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH), no long-run level
 - ▶ The variance forecast grows by $\hat{\omega}$ per day: Bitcoin from 54.1% to 108.3% over one year
 - ▶ An artefact of the unit-root estimate, not a prediction to trade on
- ▣ Long-horizon GARCH forecasts inherit all the uncertainty about $\alpha + \beta$

GARCH(1,1)-t in Six Markets

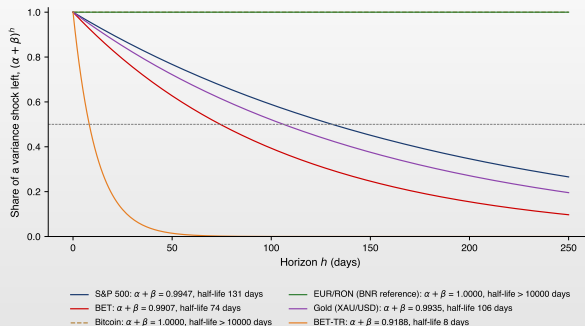
Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

- Long-run volatility in % p.a. (annualised with each series' own frequency); – when $\hat{\alpha} + \hat{\beta} \geq 0.999$
- EUR/RON: $\hat{\omega}$ is 0.0000 at four decimals (the rate moves by fractions of a ban on most days)

Interpretation: Six Volatility Personalities

- Equities (S&P 500, BET) and gold: persistence 0.99–0.995, half-lives of 3–6 months of trading days
- BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
 - ▶ BET-TR (2014–2026 only): half-life 8 days; a shorter, calmer sample gives a lower persistence
- Bitcoin: the heaviest tails ($\hat{\nu} = 3.18$: infinite fourth moment under the fitted law) and a unit root in variance
- EUR/RON: a managed exchange rate; long quiet periods, then jumps
 - ▶ 6 May 2025: +1.20% after days of $\pm 0.004\%$: standardised residual of 154

Persistence: How Fast Do Shocks Fade?



□ Share of a variance shock still present after h days, $(\hat{\alpha} + \hat{\beta})^h$; dashed line: one half

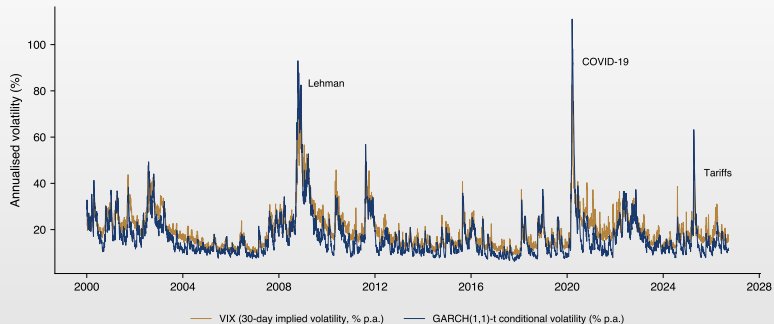
Interpretation: Persistence

- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8
- Bitcoin and EUR/RON lie on the flat line: an estimated $\alpha + \beta$ of 1 means shocks never fade in the model
- Very high persistence can be spurious: level shifts in volatility mimic a unit root [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Regime-switching alternatives: [Hamilton & Susmel \(1994\)](#)
- Practical rule: report half-lives with a confidence interval (parametric bootstrap, Seminar 5)

Delta Method for Long-Run Volatility and Half-Life

- For a smooth $g(\theta)$: $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, with $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$
- S&P 500, GARCH(1,1)-t: long-run volatility 27.3% p.a., s.e. 10.8 points: the gradient is scaled by $1/(1 - \alpha - \beta)^2$
- Half-life $h = \ln 0.5 / \ln p$, $p = \alpha + \beta$: $\partial h / \partial p = -\ln 0.5 / [p (\ln p)^2] = 24\,789$ at $\hat{p} = 0.994712$
 - ▶ $\widehat{\text{s.e.}}(\hat{p}) = 0.0047$, so $\widehat{\text{s.e.}}(\hat{h}) \approx 116$ days for $\hat{h} = 131$: a symmetric interval would include negative half-lives
- The linearisation fails because $h(p)$ explodes as $p \rightarrow 1$
 - ▶ Transform an interval for p instead (h is monotone): $[0.9855; 1.0039]$ gives $h \in [48; \infty)$
 - ▶ Or use a parametric bootstrap and a test simulated under H_0 (Seminar 5, B7)

S&P 500: GARCH Volatility and the VIX



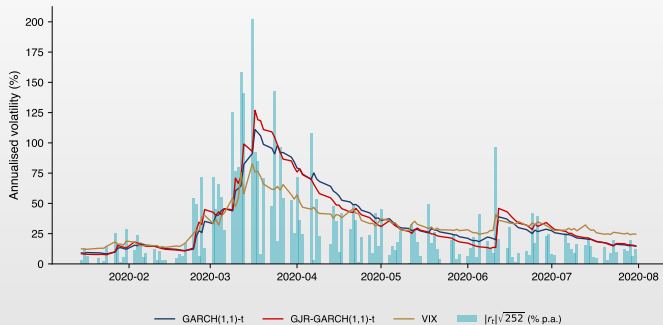
■ Annualised GARCH(1,1)-t conditional volatility $\hat{\sigma}_t \sqrt{252}$ and the VIX, the 30-day implied volatility, 2000–2026

MFM_ch5_sp500_vix

Interpretation: Statistical versus Implied Volatility

- ▣ Correlation of the two series: 0.89; GARCH uses only past returns, the VIX uses option prices
- ▣ Average VIX 19.8% against average GARCH volatility 17.0%; VIX is higher on 84% of days
 - ▶ An implied-minus-statistical gap, consistent with a **variance risk premium** (option buyers pay for insurance against volatility); the horizons also differ (30 days vs one day)
 - ▶ The premium proper compares VIX^2 with a 30-day forecast of the variance under the real-world probability
- ▣ GARCH range: from 6.2% (October 2017) to 111% (March 2020)
- ▣ GARCH reacts one day late by construction: it needs the shock before it moves

Zoom: The COVID-19 Crash, 2020



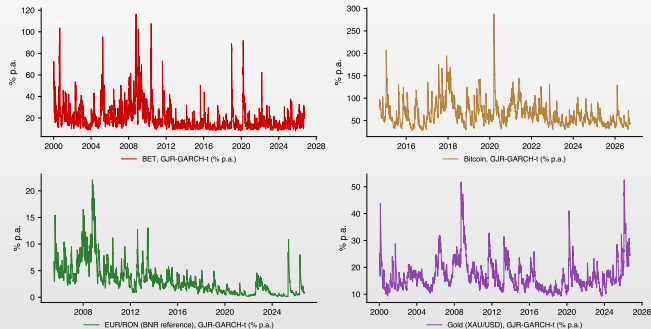
□ January–July 2020: absolute returns (annualised), GARCH-t and GJR-GARCH-t volatility, VIX

MFM_ch5_sp500_vix

Interpretation: COVID-19

- ▣ 16 March 2020: $r_t = -12.8\%$, the largest daily fall of the sample; VIX closed at 82.7%
- ▣ Peak model volatility on 17 March: GARCH-t 111%, GJR-GARCH-t 127%
 - ▶ GJR goes higher because the shocks were negative (Section 5)
- ▣ 24 February 2020 ($r_t = -3.41\%$): the VIX jumped the same day, from 17.1% to 25.0%
 - ▶ GARCH-t moved only the next day, from 12.2% to 22.5%: markets price news at once, GARCH only reacts
- ▣ After the peak, the variance forecast's deviation from its long-run level decays by the factor $\alpha + \beta$ per day; the realised path also reflects new shocks

BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t conditional volatility, annualised with each series' frequency (Bitcoin: 365 days)

Interpretation: Four Markets

- ▣ BET: the 2008 crisis dominates; short, sharp spikes afterwards; current level 26.4%
- ▣ Bitcoin: volatility of 50–100% is normal; March 2020 is the highest peak
- ▣ EUR/RON: high volatility until 2012, then a long managed calm, with bursts in 2022 and 2025
 - ▶ Volatility close to zero on many days: a daily GARCH is a poor description of a managed rate
- ▣ Gold: peaks of 52% (2008), 31% (2013), 41% (2020) and a new high of 53% in 2026

Recap: Results by Market

Key points

- ▣ One-step and multi-step forecasts follow from the recursion; the 10-day variance is a sum of daily variances
- ▣ Persistence is high everywhere; half-lives range from about a week (BET-TR) to several months
- ▣ Estimated unit roots (Bitcoin, EUR/RON) make long-run forecasts meaningless
- ▣ GARCH tracks the VIX closely but lags it; the VIX is higher on average (an implied-statistical gap, consistent with a variance risk premium)

Bad News Moves Volatility More

- ▣ Stylised fact 9 (Chapter 1): negative returns raise future volatility more than positive returns of the same size
- ▣ **Leverage effect:** a falling price raises the debt-to-equity ratio, so equity becomes riskier
 - ▶ **Volatility feedback:** higher expected volatility raises required returns, so prices fall now
- ▣ GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- ▣ Two fixes: GJR adds a term for negative shocks, EGARCH models the log-variance

GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ Glosten, Jagannathan & Runkle (1993)
 - ▶ Good news: slope α ; bad news: slope $\alpha + \gamma$; $\gamma > 0$ means asymmetry
 - ▶ Positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$; γ itself may be negative (gold, below)
- Persistence with symmetric z_t : $\alpha + \gamma/2 + \beta$ (half the shocks are negative)
 - ▶ Long-run variance $\omega/(1 - \alpha - \gamma/2 - \beta)$, finite if $\alpha + \gamma/2 + \beta < 1$; skewed z_t : replace 1/2 by $\mathbb{E}[z_t^2 1(z_t < 0)]$ (Seminar 5, A2)
- Test of symmetry: $H_0: \gamma = 0$ by a t-test or by LR against GARCH(1,1)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$

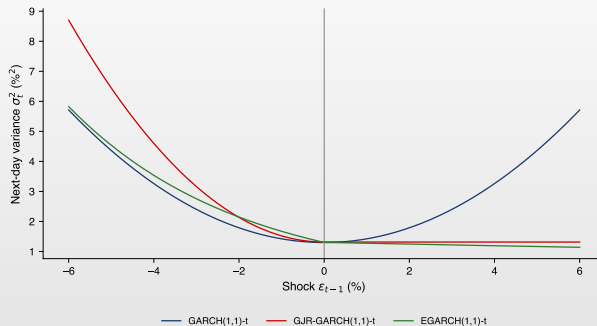
EGARCH (Nelson, 1991)

- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ Nelson (1991)
 - ▶ Size effect α , sign effect γ ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ ($t = -15.2$), $\hat{\beta} = 0.979$
- In arch, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve

- ▣ **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed at the sample variance
 - ▶ GARCH: a parabola symmetric around 0
 - ▶ GJR: two half-parabolas with different slopes
 - ▶ EGARCH: exponential in $|\varepsilon|$, tilted by γ
- ▣ It isolates how each model turns one day of news into tomorrow's variance
- ▣ Engle and Ng also propose the sign-bias tests used in the diagnostics (Section 6)

News Impact Curves for the S&P 500



- Next-day variance after a shock ε_{t-1} (in %), for the three fitted models with Student-t innovations

Interpretation: News Impact

- Starting variance 1.47 (sample variance, %²)
- A shock of -3% : GJR gives 3.16, EGARCH 2.76; a shock of $+3\%$: GJR 1.31, EGARCH 1.22
 - ▶ The variance forecast after bad news is 2.4 times that after good news in GJR, 2.3 times in EGARCH (ratios of levels, not of increases)
- Symmetric GARCH gives 2.40 for both signs: too much after good news, too little after bad news
- For the S&P 500, $\hat{\alpha} = 0$ in GJR: positive shocks do not raise volatility at all

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- All models with Student-t innovations; t-statistics with robust standard errors; $LR \sim \chi^2_1$ (critical value 3.84) if the Student-t density is correct, otherwise rely on the robust t
- Asymmetry: GJR $\hat{\gamma} > 0$ and EGARCH $\hat{\gamma} < 0$ for the same direction (bad news matters more)

Interpretation: Where Is the Leverage Effect?

- S&P 500: very strong ($LR = 235.7$); the classic leverage effect of a developed equity index
 - ▶ BET and BET-TR: present but weak ($LR = 4.9$ and 10.3)
- Bitcoin: no asymmetry detected ($LR = 0.7$); in this sample, crypto rallies are about as volatile as crashes
- Gold: **inverted** asymmetry ($\hat{\gamma}_{GJR} = -0.043$): price rises raise volatility more
 - ▶ Consistent with gold as a safe haven: it rallies in turbulent times
- EUR/RON: GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$: falls of the rate (leu appreciation) raise volatility slightly more

Recap: Asymmetry

Key points

- ▣ Symmetric GARCH ignores the sign of news; equity data reject this strongly
- ▣ GJR adds a slope γ for negative shocks; EGARCH models $\ln \sigma_t^2$ with a sign term
- ▣ The news impact curve shows the asymmetry directly
- ▣ Asymmetry is an equity phenomenon: not detected for Bitcoin, reversed for gold

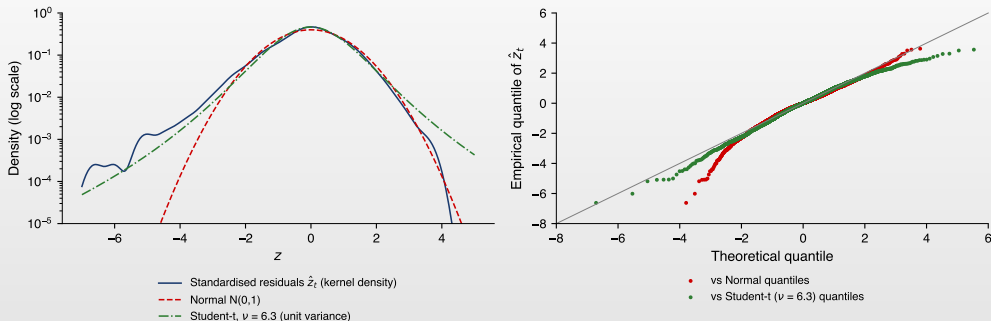
Standardised Residuals: What Should Hold

- ▣ **Standardised residual:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- ▣ Checks
 - ▶ No autocorrelation in $\hat{z}_t$: Ljung–Box $Q(10)$ on $\hat{z}_t$ (mean equation)
 - ▶ No clustering left: $Q(10)$ on $\hat{z}_t^2$ and the LM(5) test (variance equation)
 - ▶ No asymmetry left: the sign-bias tests [Engle & Ng \(1993\)](#)
 - ▶ Distribution: compare $\hat{z}_t$ with the assumed density (QQ plot)

The Sign-Bias Tests (Engle and Ng, 1993)

- Regress $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18 of Engle & Ng (1993))
- **Sign bias** (b_1): do negative shocks alone raise the next squared residual?
- **Negative / positive size bias** (b_2, b_3): does the size of negative / positive shocks matter beyond the model?
- Joint test: $nR^2 \sim \chi_3^2$ (5% critical value 7.81); it omits the gradient of the fitted variance, so it is conservative (Engle and Ng)

S&P 500: Are the Innovations Normal?



- Standardised residuals of GARCH(1,1) with Normal innovations: log-density and QQ plot against the Normal and Student-t distributions

Interpretation: The Tails After GARCH

- Excess kurtosis drops from 10.7 (returns) to 1.73 (standardised residuals): GARCH explains most of it
- But not all: Jarque–Bera 1153; skewness -0.53
 - ▶ 15 residuals below -4 ; the Normal distribution expects 0.21, a Student-t with $\nu = 6.3$ expects 8.5
 - ▶ Largest in absolute value: $\hat{z} = -6.6$ on 27 February 2007, a large fall in a calm period
- The left tail is longer than the right: skewed-t with $\hat{\lambda} = -0.150$ (GJR, $\hat{\eta} = 7.7$)
- Hence BIC prefers skewed-t innovations for the S&P 500

S&P 500: Diagnostics of Five Models

Model	$Q(10)$ on $\hat{z}$	$Q(10)$ on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	3.64	41.8
GARCH-t	21.9	13.6	5.8	3.56	40.7
GJR-t	22.3	10.2	4.5	3.61	28.1
EGARCH-t	27.6	10.5	8.2	2.04	9.9
GJR-skewed-t	19.7	10.7	5.0	3.91	31.8

- 5% critical values: χ^2_{10} : 18.3; χ^2_5 : 11.1; χ^2_3 : 7.81; $|t| > 1.96$
- Before modelling, $Q(10)$ on r_t^2 was 6133

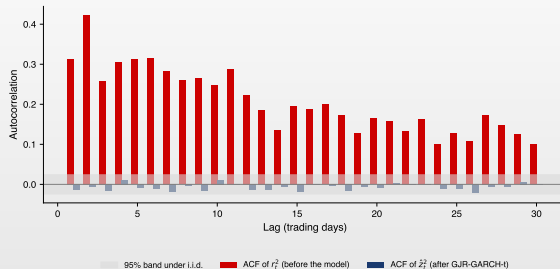
Interpretation: What the Diagnostics Say

- ▣ The naive tests do not detect remaining clustering: $Q(10)$ on $\hat{z}^2$ falls from 6133 to 10–16, $LM(5)$ is not significant; parameter-adjusted diagnostics follow
- ▣ Symmetric GARCH fails the sign-bias test (joint 40.7): asymmetry left over
 - ▶ GJR still fails (28.1); EGARCH comes closest (9.9, $p = 0.020$)
- ▣ $Q(10)$ on $\hat{z}$ around 20–28: some autocorrelation in the mean, which a constant mean ignores
- ▣ With $n = 6\,714$, small effects are significant: judge economic size, not only p-values

Residual Diagnostics Have Estimated Parameters

- $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ_m^2 Li & Mak (1994)
 - ▶ Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n} \hat{r} \rightarrow N(0, C)$, $C = I - H' J^{-1} H / (\kappa_z - 1)$
 - ▶ $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$: how the fitted variance absorbs lag- k dependence; the ARMA rule χ_{m-p-q}^2 does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
 - ▶ The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence
- $Q(10)$ on $\hat{z}_t^2$: Ljung–Box 16.4 ($p = 0.089$); corrected $n \hat{r}' C^{-1} \hat{r} = 19.1$ ($p = 0.038$): the 5% verdict flips
- The naive test is conservative: it under-rejects a misspecified variance equation; the joint sign-bias test carries the same caveat

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500: autocorrelations of r_t^2 and of $\hat{z}_t^2$ from GJR-GARCH-t, lags 1–30
- After the model, the autocorrelations sit inside the i.i.d. band: no remaining variance dynamics are detected at these lags

Diagnostics in Other Markets

- BET: $Q(10)$ on $\hat{z} = 116.6$: strong autocorrelation in the mean (thin trading, Chapter 2)
 - ▶ Remedy: an AR(1) mean equation, $r_t = \mu + \phi r_{t-1} + \varepsilon_t$
- Bitcoin: $Q(10)$ on $\hat{z}^2 = 4.7$ and the joint sign-bias statistic 1.3 do not reject
 - ▶ Under the fitted Student-t ($\hat{\nu} = 3.18 < 4$), $\text{Var}(z_t^2) = \infty$: the χ^2 calibration of tests on $\hat{z}_t^2$ is not justified; also check $|\hat{z}_t|$
- EUR/RON: residual excess kurtosis 3190, driven by the single residual of 154 (May 2025)
 - ▶ One observation can dominate the diagnostics: always look at the largest residuals
- Gold: EGARCH leaves clustering ($Q(10)$ on $\hat{z}^2 = 48.8$) although its BIC is among the best

Recap: Innovations and Diagnostics

Key points

- ▣ Standardised residuals must look i.i.d.: test their level, their squares and their signs
- ▣ GARCH removes the clustering and most of the kurtosis, not the skewness
- ▣ Heavy-tailed, skewed innovations and asymmetric variance are both needed for the S&P 500
- ▣ Diagnostics can disagree with information criteria; one extreme residual can drive them

The Proxy Problem

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g. r_t^2
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
 - ▶ Its R^2 against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
 - ▶ S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
 - ▶ Robust: MSE $(r_t^2 - h_t)^2$ and **QLIKE** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast
 - ▶ Not robust: losses on volatility $|r_t| - \sqrt{h_t}$ or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

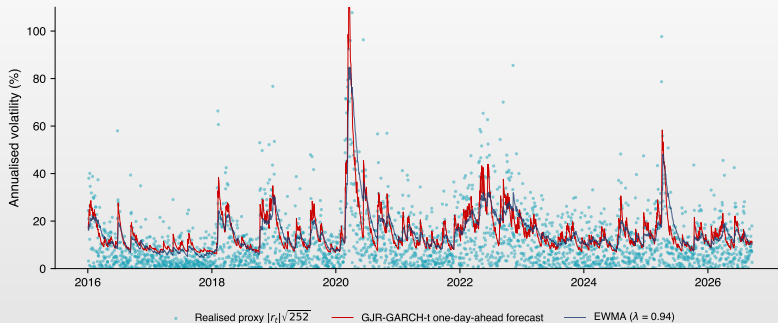
Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$
 - ▶ Unbiased forecast: $a = 0$, $b = 1$ (joint Wald test with HAC errors); R^2 measures information content
 - ▶ $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
 - ▶ $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$ with a HAC standard error; $\approx N(0, 1)$ under equal accuracy
 - ▶ $DM < -1.96$: model A significantly better

Out-of-Sample Design

- ▣ S&P 500; forecast period 2016–2026: 2 692 one-day-ahead forecasts
- ▣ Parameters re-estimated at the start of each year on all data since 2000 (expanding window)
 - ▶ Within the year, the variance is updated daily with fixed parameters: no look-ahead
- ▣ Models: GARCH with Normal and Student-t innovations, GJR-t, EGARCH-t
- ▣ Benchmarks: EWMA ($\lambda = 0.94$) and the 20-day historical variance (Hist-20)
- ▣ Also 22-day (one-month) cumulative variance forecasts on 122 non-overlapping windows

Forecasts versus Realised Moves, 2016–2026



One-day-ahead volatility forecasts (annualised) of GJR-GARCH-t and EWMA; teal dots: $|r_t|\sqrt{252}$

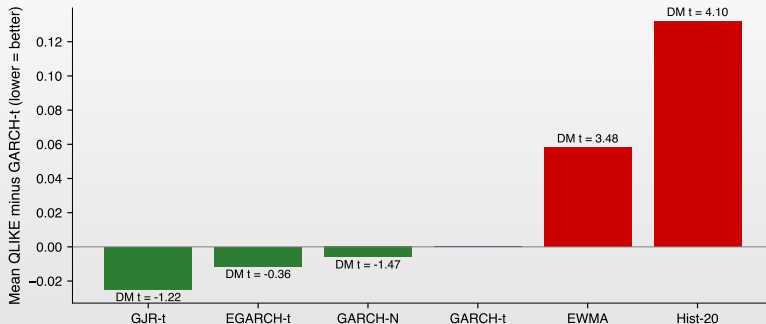
S&P 500: Forecast Evaluation, 2016–2026

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	-1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	—	0.88	0.238	0.80
GJR-t	0.732	21.8	-1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	-0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ: $r_t^2 = a + b h_t$; Wald p: test of $a = 0$, $b = 1$

 MFM_ch5_forecast_evaluation

QLIKE Relative to GARCH-t



- Mean QLIKE of each model minus that of GARCH-t, with the Diebold–Mariano t-statistic

Interpretation: Does Anything Beat GARCH(1,1)?

- Best QLIKE: GJR-t (0.732); but the GARCH-family differences are not significant (DM t from -1.22 to -0.36)
- EWMA and Hist-20 are significantly worse (DM t = 3.48 and 4.10)
 - ▶ Mean reversion and estimated parameters pay off
- MZ: unbiasedness is not rejected for GARCH-type forecasts (b close to 1, Wald $p > 0.6$); it is rejected for Hist-20 ($p = 0.03$)
- R^2 of only 0.15–0.29: the noise of r_t^2 , not a bad model
- Same message as [Hansen & Lunde \(2005\)](#): GARCH(1,1) is hard to beat; asymmetry helps a little for equities

Comparing Forecasts from Estimated, Nested Models

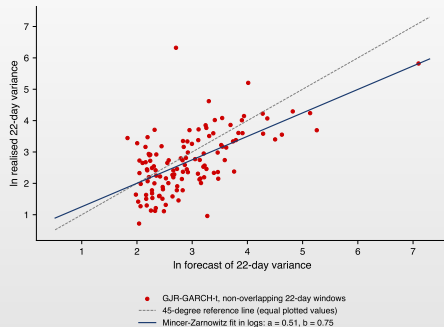
- DM treats forecasts as given; ours come from estimated parameters, and GARCH-t is nested in GJR-t
 - ▶ Expanding window, nested models: under H_0 the loss differential degenerates, so DM is not $N(0, 1)$ [West \(1996\)](#), [Clark & McCracken \(2001\)](#)
- [Giacomini & White \(2006\)](#): test the forecasting **method** (model + rolling window of fixed length W)
 - ▶ Valid for nested models and estimated parameters, because estimation noise does not vanish
 - ▶ Conditional test $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = 0$ with instruments $h_t = (1, d_t)$: $GW = n \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2_2$, $Z_{t+1} = h_t d_{t+1}$
- Six models and five pairwise DM tests: a multiple-comparison problem
 - ▶ **Model Confidence Set** [Hansen, Lunde & Nason \(2011\)](#): drop the worst model until equal accuracy is not rejected; the set contains the best model with probability $\geq 1 - \alpha$

S&P 500: Giacomini–White Tests and the Model Confidence Set

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.751	1.87	5.85	0.054	0.586
GARCH-t	0.743	—	—	—	0.353
GJR-t	0.738	-0.18	2.49	0.288	1.000
EGARCH-t	0.746	0.07	3.21	0.201	0.586
EWMA	0.815	3.20	11.40	0.003	0.008
Hist-20	0.889	3.87	16.41	< 0.001	0.001

- Rolling window $W = 1\,000$ days, re-estimated each January, 2 692 forecasts; tests against GARCH-t, HAC with 5 lags
- MCS on the expanding-window forecasts of Section 7: QLIKE, statistic T_R , stationary bootstrap (mean block 10, 10 000 draws)
- 90% MCS: GARCH-N, GARCH-t, GJR-t, EGARCH-t; EWMA and Hist-20 excluded: the same verdict as DM, now with a controlled error rate

One-Month Forecasts: Mincer–Zarnowitz in Logs



- GJR-GARCH-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows

Interpretation: Monthly Horizon

- ▣ Aggregating 22 days reduces the proxy noise: R^2 in logs 0.38 (GJR-t); not comparable with 0.28 for one day in levels (other horizon, other scale)
- ▣ Slope in logs $b = 0.75$ (s.e. 0.07): a descriptive summary, not a test of unbiasedness
 - ▶ $\mathbb{E}[\ln Y \mid \mathcal{F}] \neq \ln \mathbb{E}[Y \mid \mathcal{F}]$: an unbiased variance forecast need not give $b = 1$ in logs, so $b < 1$ does not prove over-dispersed forecasts
- ▣ In levels the slope is only 0.27: two COVID-19 windows (February and March 2020) dominate the regression
- ▣ Use QLIKE for proxy-robust forecast comparisons [Patton \(2011\)](#); use log plots descriptively when a few crisis months carry huge variances

Recap: Forecasting

Key points

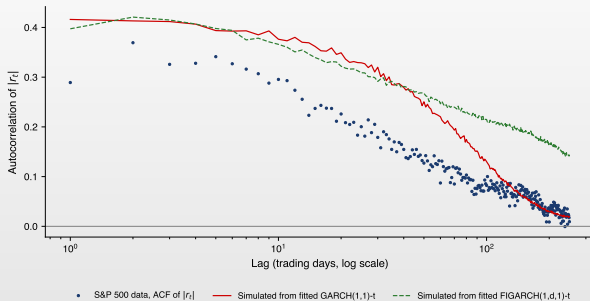
- ▣ Evaluate forecasts out of sample, with parameters estimated only on past data
- ▣ With a noisy proxy use robust losses (QLIKE, MSE) and the Diebold–Mariano test
- ▣ Nested models with estimated parameters: Giacomini–White with a rolling window; many models: Model Confidence Set
- ▣ Mincer–Zarnowitz checks unbiasedness; low R^2 at one day is expected
- ▣ GARCH beats EWMA and historical variance; members of the GARCH family are hard to separate

GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987): $r_t = c + \kappa \sigma_t + \varepsilon_t$ (or $\kappa \sigma_t^2$)
 - ▶ $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t ; 0.028 (t = 2.35) with σ_t^2
 - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

 MFM_ch5_long_memory

Long Memory: Where GARCH Falls Short



- S&P 500: autocorrelation of $|r_t|$ (lags 1–250, log scale) in the data and in long paths simulated from the fitted GARCH(1,1)-t and FIGARCH(1,d,1)-t

Interpretation: Long Memory

- Data: 0.289 at lag 1, 0.082 at lag 100, 0.009 at lag 250: a slow, roughly log-linear decline
 - ▶ GARCH: too high at short lags (0.416), then an exponential collapse: its ACF has a characteristic time
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996): fractional integration $(1 - L)^d$ gives hyperbolic decay
 - ▶ $\hat{d} = 0.60$ (s.e. 0.06); BIC 34.2 lower than GARCH, but the simulated memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: Chapter 9 uses realised volatility and the HAR model

Other Limits of GARCH

- ▣ **Structural breaks:** shifts in the volatility level inflate $\alpha + \beta$ towards 1 [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Our BET-TR (2014–2026) persistence 0.9188 vs BET (2000–2026) 0.9907: the sample matters
- ▣ **Reaction lag:** GARCH responds after the shock; implied volatility, e.g. the VIX, moves before
- ▣ **Daily data only:** intraday information is lost; realised measures use it (Chapter 9)
- ▣ **One asset at a time:** portfolios need covariances; multivariate GARCH and DCC in Chapter 6
- ▣ **Managed prices** (EUR/RON): a daily GARCH describes interventions poorly

Beyond GARCH(1,1): Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma_{t+1}^2 = \omega + A s_t + B \sigma_t^2$, s_t the scaled score of the Student-t density
 - ▶ The score is bounded in z_t : an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ_t^2 to a realised variance
 - ▶ Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma_t^2 = \tau_t g_t$, a slow long-run component τ_t driven by macro data (mixed frequencies)
 - ▶ A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1
- Each extension is judged like GARCH: out of sample, robust loss, Model Confidence Set

Beyond Variance: Entropy and Tail Risk

- [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
 - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): the entropy of intraday Bitcoin returns proposed as an additional signal for daily Value at Risk
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- GARCH volatility feeds directly into Value at Risk and expected shortfall: Chapters 7–8

An Open Question

- Do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample, once the number of models compared is accounted for?
- The question is live
 - ▶ Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models for the realised variance of Dow Jones stocks
 - ▶ Hansen, Lunde & Nason (2011): the Model Confidence Set keeps every model not significantly worse than the best, the right object when many models compete
 - ▶ This chapter: within the GARCH family, the DM statistic ranges from -1.22 to -0.36 : no significant winner
- Formal statement for models A and B, QLIKE loss L
 - ▶ $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$; $H_0: \mathbb{E}[d_t] = 0$; $H_1: \mathbb{E}[d_t] < 0$; DM statistic with HAC errors Diebold & Mariano (1995)
 - ▶ Falsified if A leaves the Model Confidence Set, or if the sign of $\bar{d}$ flips across pre-registered subperiods

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
 - ▶ Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy [Patton \(2011\)](#)
- **Hypotheses:** *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code:** *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from data/market and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
 - ▶ Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM $t = -1.22$ (Section 7)

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *“Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction.”*
 - ▶ Placebo: random splits of the same sizes; the gap in $\bar{d}$ between down-day and up-day subsets must exceed the 95% band of random gaps
- **Critique:** *“Act as a referee: could the proxy r_t^2 , one crisis year or the HAC lag create my result? Give the test that would expose each.”*
- **Report:** all splits and losses tried, adjusted p -values, code, seeds and package versions
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Look-ahead:** parameters fitted on the full sample, or h_t built with r_t itself; h_t uses data up to the close of day $t - 1$ and is evaluated against $r_t^2 \Rightarrow$ assert the last estimation date precedes each origin; change r_t and check that h_t does not move
- **Non-robust loss:** losses on $|r_t|$ or on logs can rank forecasts wrongly with a noisy proxy [Patton \(2011\)](#) $\Rightarrow$ QLIKE and MSE only
- **Model mining:** 3 variance equations $\times$ 4 distributions $\times$ windows $\Rightarrow$ Model Confidence Set or Bonferroni over the pre-registered set
- **Influential periods:** a crisis year can dominate $\bar{d} \Rightarrow$ report subperiods, and state in advance whether any year may be excluded
- **Scale and convergence:** returns in % vs decimals, failed optimisations $\Rightarrow$ check the optimiser flags and one variance update step by step
- **Invented references or functions:** unresolvable DOIs, arch arguments that do not exist $\Rightarrow$ Crossref and running the code; log in `AI_ERRORS.md`

Mini-Case: Does Asymmetry Improve the Forecast?

- A typical assistant proposal: "GJR-t beats GARCH-t because of the leverage effect"; $d_t = \text{QLIKE}(\text{GJR-t}) - \text{QLIKE}(\text{GARCH-t})$
- Replicate: 2 692 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$ ($p = 0.22$): not significant
- Testable implication of leverage: the gain sits after down days
 - ▶ After $r_{t-1} < 0$ (1 223 days): $\bar{d} = -0.015$, $t = -0.51$; after $r_{t-1} \geq 0$ (1 469): $\bar{d} = -0.034$, $t = -1.12$
 - ▶ Not supported: GJR has $\hat{\alpha} = 0.000$, so a rise does not raise its forecast; the gain is no smaller after up days
- Subperiods: 2016–2020 $t = -0.02$; 2021–2026 $t = -4.06$ ($p < 0.001$); without 2020 $t = -4.26$
 - ▶ The 2021–2026 gain survives Bonferroni over the 5 splits ($0.05/5 = 0.01$); dropping 2020 is legitimate only if pre-registered
- Interpretation: asymmetry helps, but not through the channel proposed, and not in 2016–2020

Your Turn: A Team-Project Seed

- Question: do machine-learning forecasts on daily data enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
 - ▶ Realised-variance (HAR) branch: only the SPY ETF, an S&P 500 proxy with 5-minute bars in data/intraday from October 2020, evaluated on 2021–2026; there are no intraday BET data
- Replicate first: the QLIKE table of Section 7 (GARCH- t 0.757, EWMA DM $t = 3.48$) and the mini-case splits
- Deliverables
 - ▶ A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
 - ▶ Code on data/market (daily) and data/intraday (SPY), the Model Confidence Set at 10%, a referee-style self-critique
- Log every prompt, tool and accepted output in AI_USE.md; every error caught (look-ahead, wrong scale, invented DOI) and the test that caught it in AI_ERRORS.md
- Code of the mini-case: [Quantlets/Ch_05/ai_discovery_case.py](#)

Key Takeaways

What to remember

- ▣ Returns are nearly unpredictable, their size is not: GARCH turns clustering into forecasts
- ▣ GARCH(1,1): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta$ close to 1 in all our markets
- ▣ Equity volatility is asymmetric (GJR, EGARCH); for Bitcoin no asymmetry is detected; gold is reversed
- ▣ Innovations are heavy-tailed and skewed: Student-t or skewed-t, with robust standard errors
- ▣ Out of sample, GARCH models beat EWMA and historical variance; among themselves, differences are small

A Practical Checklist

- ▣ 1. Test for ARCH effects: Q on r_t^2 and the LM test
- ▣ 2. Fit GARCH, GJR and EGARCH with Normal, Student-t and skewed-t innovations; returns in %
- ▣ 3. Check convergence, bounds and robust standard errors (Gaussian QMLE: valid only if $\mathbb{E}z_t^4 < \infty$); boundary-adjusted LR and BIC
- ▣ 4. Diagnose the standardised residuals (level, squares, signs, largest values)
- ▣ 5. Evaluate forecasts out of sample: QLIKE, Diebold–Mariano or Giacomini–White, Model Confidence Set, Mincer–Zarnowitz
- ▣ 6. Report the half-life with its uncertainty; distrust long-run numbers when $\alpha + \beta \approx 1$

Further Resources

▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

▣ Code

- ▶ All charts and tables: [Quantlets/Ch_05](#); Python package [Sheppard, arch](#)

Looking Ahead

- ▣ Chapter 6: dependence between markets, multivariate volatility and dynamic correlations
- ▣ Chapter 7: Value at Risk and expected shortfall, with GARCH volatility as an input
- ▣ Chapter 8: backtesting the risk forecasts
- ▣ Chapter 9: realised volatility and long memory
- ▣ Seminar 5: multi-step variance, delta method and robust versus classical QMLE variance, boundary LR, Giacomini–White and MCS, IGARCH test by bootstrap, US spillovers to the BET

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